A-A-53862
20 March 1989

COMMERCIAL ITEM DESCRIPTION

POLISHERS, DENTAL; MANDRELS, DENTAL HANDPIECE

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers six unmounted polishers and two dental handpiece mandrels.

Salient characteristics.

General.

The polishers shall be suitable for use as a pre-polisher and a final finisher of amalgam, gold, other cast alloy restorations, and composite restorations, as specified herein. The mandrels shall be suitable to securely hold the polisher on one end and attaching to the dental handpiece with the other end. The force required to snap polishers on and off shall be 3-15 pounds. Polishing shall be accomplished with minimal pressure. The grinding shall be virtually vibration free. The polishers, Type I, shall be of the following classes and shapes. The mandrels, Type II, shall be of the following styles:

TYPE I Polishers

Class 1. Pre-polisher amalgam
Shape 1. Cup
Shape 2. Point

Class 2. Finishing amalgam
Shape 1. Cup
Shape 2. Point

Class 3. Finishing composite
Shape 1. Cup
Shape 2. Point

TYPE II Mandrels

Style 1. Latch-type angle handpiece
Style 2. Friction grip angle handpiece

ANMC N/A FSC 6520

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The polishers shall be suitable for use wet and at speeds of 10,000 rpm or less. The polishers shall supplement each other in their use.

Material. The polishers shall be fabricated of flexible, molded polysiloxane rubber impregnated with suitable abrasive agents and color coding material. The abrasive and color coding material shall be uniformly distributed. Color shall be homogeneous. Abrasive shall be free of dirt, mold release and other contaminants. The mandrels shall be fabricated from Class 3, Type 303 austenitic stainless steel in accordance with ASTM F899.

Style, design and dimensions. Type I shall be polishers. Polishers shall be so molded as to provide a snug snap-on fit to the mandrel, Type II.

The Class 1 polishers shall be unmounted, snap-on type and brown in color to indicate their suitability for use as a pre-polisher of amalgam, gold, and other cast alloy restorations. Total indicated runout shall be .015 maximum when tested as specified in test section.

The Class 2 polishers shall be unmounted, snap-on style and green in color to indicate their suitability for use in the final finishing of amalgam gold, and other cast alloy restorations. Total indicated runout shall be .015 maximum when tested as specified in test section.

Class 3 polishers shall be unmounted, snap-on style and yellow in color to indicate their suitability for use in finishing composite restorations. Total indicated runout shall be .015 maximum when tested as specified in test section.

Shape 1 shall be polishing cups. The polishing cups shall have dimensions as specified in Figure 1.

Shape 2 shall be polishing points. The polishing points shall have dimensions as specified in Figure 1.

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Type II shall be mandrels. Style 1 mandrel shall be suitable for use in a latch-type angle handpiece. Style 2 mandrel shall be suitable for use in a friction grip angle handpiece. The distal configuration of the mandrels shall consist of an indented annular portion for snap-on polishers, Type I, followed by a tongue portion (widened and flattened) that resists rotational torque. Dimensions of mandrels shall be as specified in Figure 2, for Style 1 and Figure 3 for Style 2. Shank dimensions, except overall length shall be in accordance with Figure 1 for Style 1 and Figure 4, class 4 for style 2 of ANSI/ADA Specification No. 23 for dental excavating burs.

The mandrels and polishers shall be suitable for use, with each other, as applicable.

Finish. The mandrels shall have a polished finish, or as an option, the finish shall be black electroplated. Style 2 shank surface roughness shall have a RMS value of not less than 63 micro-inches for the polished finish.

Corrosion resistance. The mandrels shall show no signs of corrosion when tested in accordance with the boil test of ASTM 1089.

Workmanship. Polishers shall be smooth and free from large nicks, cracks, voids, excess flash, hardening, inflexibility and brittleness. The mandrels shall be free from cracks, pits, burrs, sharp edges or rough areas. Polishers and mandrels shall be free from defects which detract from their appearance or impair their serviceability.

Unit. Package (PG). One package containing 30 polishers or one package containing 10 mandrels, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests to include dimensions, shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Runout. The polisher shall be placed and held in a "V" block or precision chuck, and rotated slowly. A dial indicator sensitive to 0.0002 inch and equipped with a knife edged foot, or a shadow-graph or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, the indicator shall be mounted with the shaft perpendicular to the axis of the polisher, and so positioned that the foot contacts the polisher at the point of smallest diameter. The maximum and minimum readings obtained during one complete 360-degree rotation of the polisher shall be measured and recorded to the nearest 0.0005 inch. The difference between these two readings shall be taken as the total indicated runout.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be package of mandrels or polishers, as specified. Mandrels and polisher shall be examined for defects. Defects shall include, but not limited to those listed below.

Defects

Color of polisher not as specified.
Polisher is hardened, inflexible or brittle.
Mandrels not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
Finish of the mandrels not as specified.
Type, Class, Style and Shape not as specified.

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Sampling for tests. Sampling for tests to include dimensions, shall be conducted in accordance with MIL-STD-105 and as specified herein. The unit of product for test purposes shall be one package as specified. AQL shall be 1.0 (percent defective) with an inspection level of S-2.

Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION	TYPE	CLASS	SHAPE	STYLE
6520-01-089-5660	POLISHER, DENTAL, Cup, Brown, Unmounted, Snap-on Type, 30s.	I	1	1	--
6520-01-089-5661	POLISHER, DENTAL, Point, Brown, Unmounted, Snap-on Type, 30s.	I	1	2	--
6520-01-089-5662	POLISHER, DENTAL, Cup, Green, Unmounted, Snap-on Type, 30s.	I	2	1	--
6520-01-089-5663	POLISHER, DENTAL, Point, Green, Unmounted, Snap-on, Type, 30s.	I	2	2	--
6520-01-089-5664	POLISHER, DENTAL, Cup, Yellow, Unmounted, Snap-on Type, 30s.	I	3	1	--
6520-01-089-5665	POLISHER, DENTAL, Point, Yellow, Unmounted, Snap-on Type, 30s.	I	3	2	--
6520-01-089-5670	MANDREL, DENTAL HANDPIECE, Angle Handpiece, Latch-type, 10s	II	--	--	1
6520-01-089-5671	MANDREL, DENTAL HANDPIECE, Friction Grip Angle Handpiece, 10s.	II	--	--	2

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

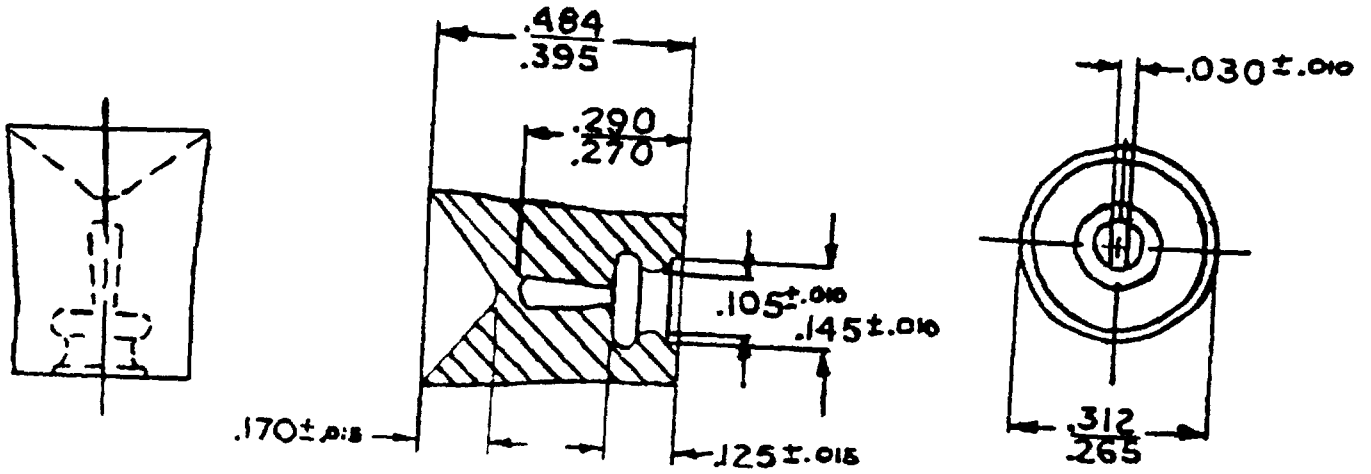
DLA-DM

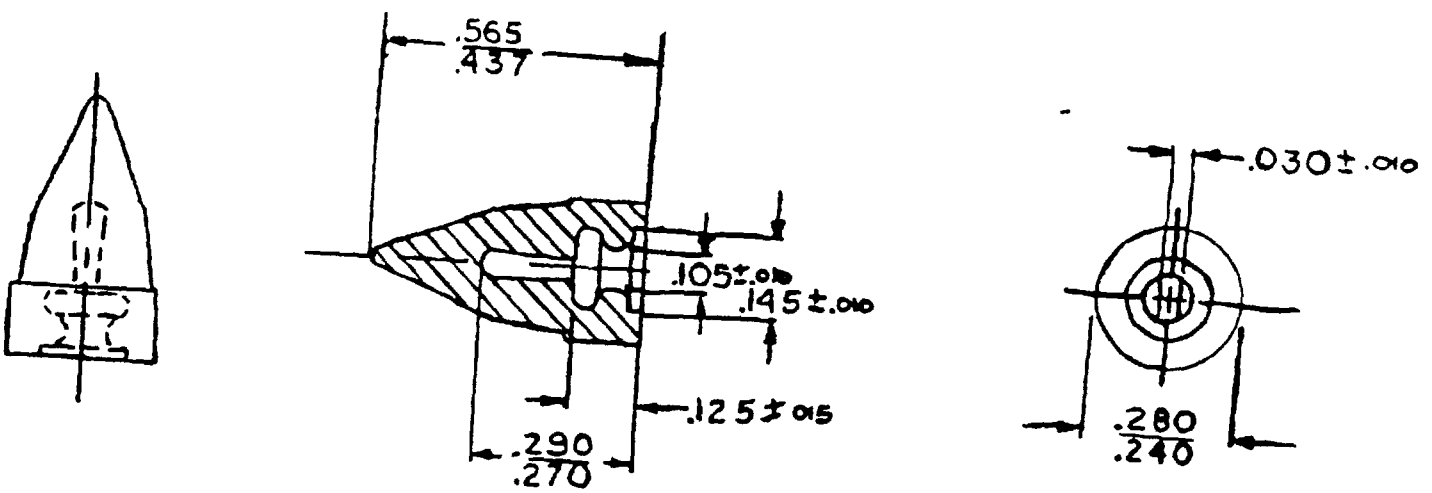
CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSSs
PHS
FDA-MPQAS

Project No. 6520-2187

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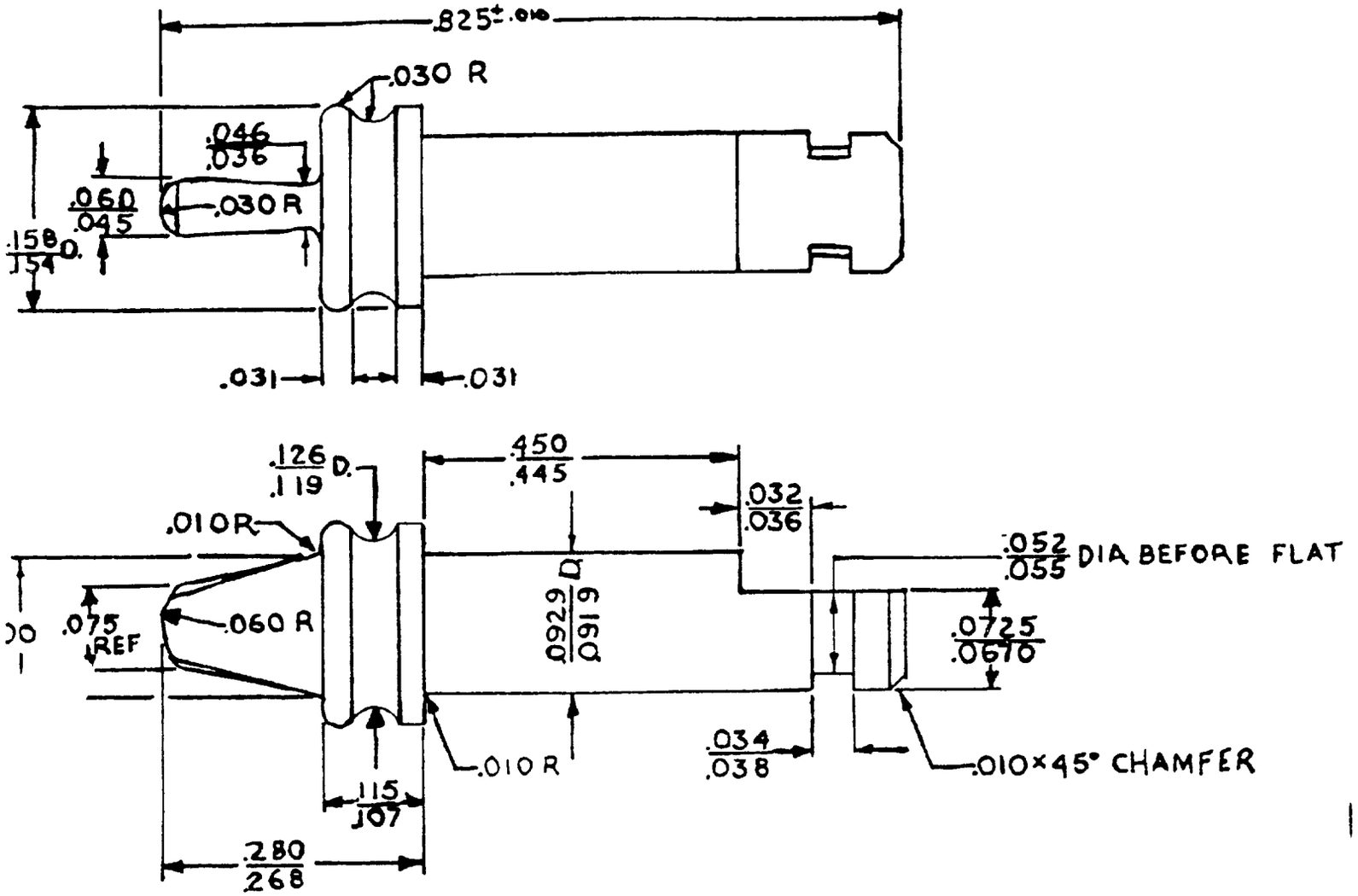
TYPE I - SHAPE 1 POLISHING CUPS

TYPE II - SHAPE 2 POLISHING POINTS

ALL DIMENSIONS ARE IN INCHES

FIGURE 1

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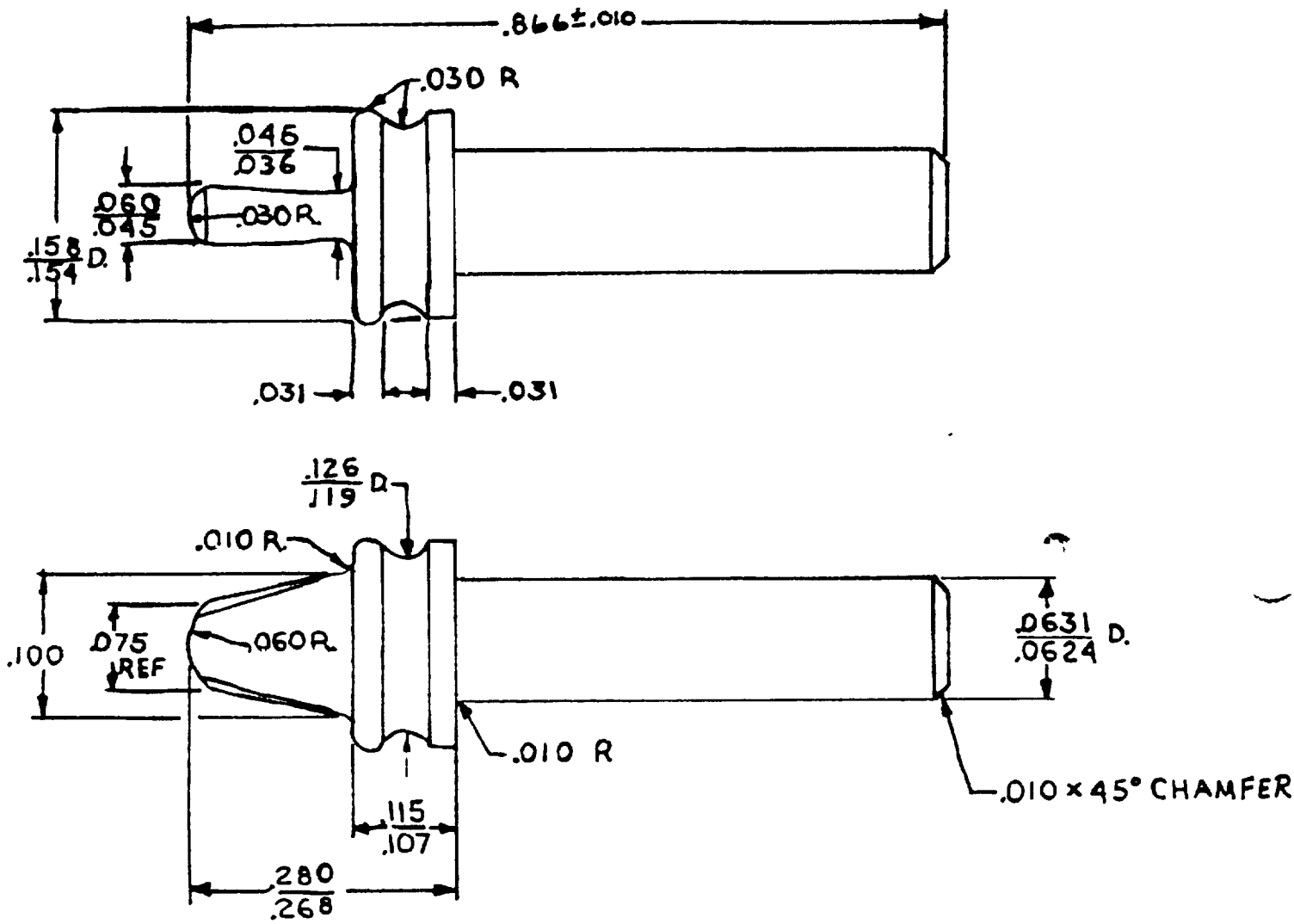
NOTE:
TOLERANCE ±.005 UNLESS SPECIFIED OTHERWISE

ALL DIMENSIONS IN INCHES.

STYLE 1 - LATCH-TYPE MANDREL

FIG RE 2

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NOTE:
TOLERANCE ±.005 UNLESS SPECIFIED OTHERWISE

ALL DIMENSIONS IN INCHES.

STYLE 2 FRICTION GRIP MANDREL

FIGURE 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
4-A-53862	POLISHERS, DENTAL; MANDRELS, DENTAL HANDPIECE

NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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